FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Advanced)-2021

PAPER –2

TEST DATE: 30-05-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total **54** questions.
- Each subject (PCM) has **18** questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section – B & Section-C.**

Section-A (01 – 06, 19 – 24, 37– 42) *this section contains* ***18 multiple*** *choice questions.*
Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
Negative Marks **: –2** In all other cases

Section-B (07 – 12, 25 – 30, 43– 48) *contains* ***18 Numerical*** *based questions with* ***Single digit integer*** *as answer, ranging from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-C (13 – 18, 31 – 36, 49– 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

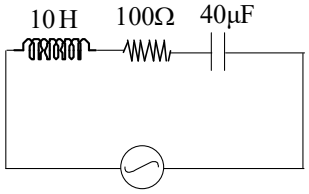
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. If the effective voltage of the AC source is 200 V and its frequency is variable:



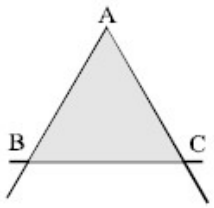
(A) The frequency at which power dissipated in resistor is maximum is 50 Hz.

(B) The frequency at which power dissipated in resistor is maximum is $25/\pi$ Hz.

(C) The frequency at which power dissipated across the resistor drops to half's maximum value is $45/2\pi$ Hz.

(D) The frequency at which power dissipated across the resistor drops to half's maximum value is $20/\pi$ Hz.

2. A soap film is formed on a vertical equilateral triangular frame ABC. Side BC can move vertically always remaining horizontal. If m is the mass of rod BC, and T is surface tension of soap film. Mark the correct statement(s):-



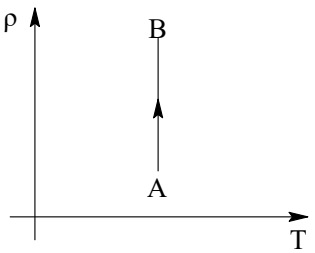
(A) The length of BC in equilibrium is $\frac{mg}{2T}$

(B) The length of BC in equilibrium is $\frac{mg}{T}$

(C) The time period of small oscillations about the equilibrium position is $\pi\sqrt{\frac{\sqrt{3}m}{T}}$

(D) The time period of small oscillations about the equilibrium position is $\pi\sqrt{\frac{m}{\sqrt{3}T}}$

3. The density (ρ) of an ideal gas varies with temperature T as shown in figure. Then:



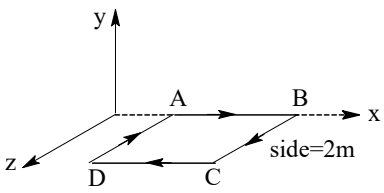
(A) The product of P & V at A is equal to product of P & V at B

(B) Pressure at B is greater than the pressure at A

(C) Work done by the gas during the process AB is negative.

(D) The change in internal energy from A to B is zero.

4. A square wire frame is hinged about one of its sides AB. Which of the following magnetic fields can hold it in equilibrium in horizontal position shown. It has a mass of 500 gm:



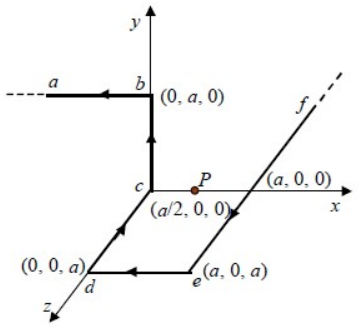
(A) $3\hat{i}+\frac{5}{2}\hat{k}$

(B) $\frac{5}{2}\hat{i}-3\hat{j}+2\hat{k}$

(C) $2\hat{j}-\frac{5}{2}\hat{k}$

(D) $\hat{i}+\frac{5}{2}\hat{j}+\frac{5}{2}\hat{k}$

5. A very long wire $abcdef$ carrying current has the shape as shown in the figure. The magnetic field at point P in terms of unit vectors.



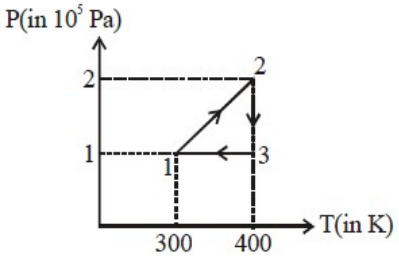
(A) Due to wire ab is $\frac{\mu_0 I}{4\pi a}\left[1-\frac{1}{\sqrt{5}}\right]\hat{k}$

(B) Due to wire bc is $\frac{-\mu_0 I}{\pi a\sqrt{5}}\hat{k}$

(C) Due to wire cd is $\frac{-\mu_0 I}{\pi t\sqrt{5}}\hat{j}$

(D) Due to wire of ef is $\frac{-\mu_0 I}{2\pi t}\left[1+\frac{2}{\sqrt{5}}\right]\hat{j}$

6. A certain mass of oxygen, treated like an ideal gas undergoes a cycle shown on P-T diagram here. The maximum volume of gas during the cycle is 16 litres. Select the correct statement(s):
(Take: R = 25/3 J/mol K)



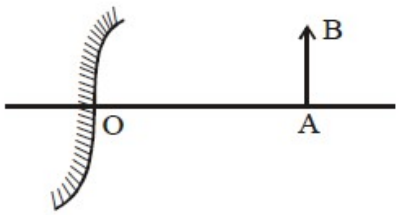
(A) The mass of gas is 15.36 gm
(B) The mass of gas is 30.72 gm
(C) The minimum volume of gas during cycle is 12 litres
(D) The minimum volume of gas during cycle is 8 litres.

SECTION – B
(Single Digit Integer Type)

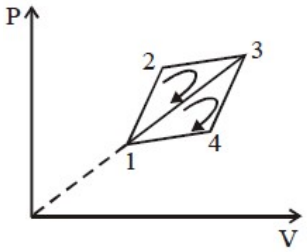
This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

7. A lizard of mass 4g is warming itself in the bright sunlight. Its surface area is $4.2\,cm^2$, but its body casts a shadow of area $1.4\,cm^2$ on a piece of paper held perpendicular to sun rays. The intensity of sun rays at earth is $1000\,W/m^2$. Assume that specific heat capacity of lizard is same as that of water and its body behaves like a black body, also neglect the radiation from the lizard's body. How long (in minutes) must lizard lie in sun in order to raise its temperature by $5°C$. Specific heat capacity of water is $4200\,J/kg°C$. If your answer is n fill value $\frac{n}{5}$.

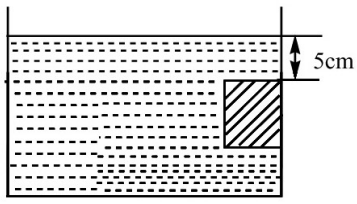
8. A Composite mirror is formed by arranging the upper half of a concave mirror above the lower half of a convex mirror as shown. A real linear object AB is placed on the common principal axis and it is found that images formed by the two halves of the composite mirror are equidistant from the point O. If the radii of curvature of the concave mirror and convex mirror halves are 20 cm and 30 cm respectively. The distance of the object AB from O is given by x cm then $\frac{x}{10}$ is:



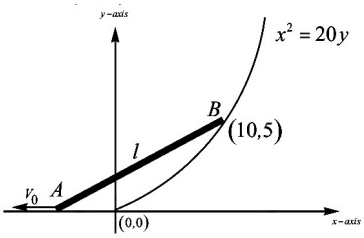
9. The efficiency of a thermodynamic cycle $1\text{-}2\text{-}3\text{-}1$ (see picture) is 20% and for another thermodynamic cycle $1\text{-}3\text{-}4\text{-}1$ efficiency is equal to 10%. Determine the efficiency $\eta(\text{in }\%)$ of the thermodynamic cycle $1\text{-}2\text{-}3\text{-}4\text{-}1$. The gas is assumed to be ideal. Fill $\eta/4$ in OMR sheet:



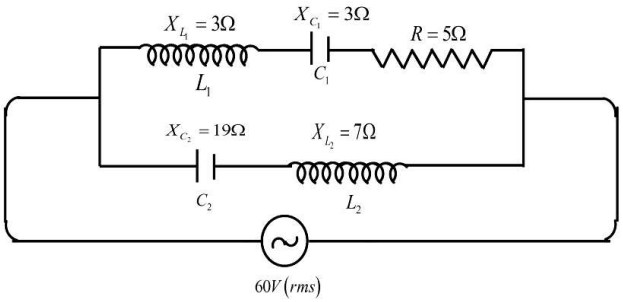
10. A cube of side *a* = 10cm is welded to the side wall of container so that the upper surface of the cube lies 5 cm below the free surface of water. Specific gravity of the cube is 2. If net contact force between the wall and the cube is $\left(10\sqrt{x}\right)N$, find the value of x, $g = 10m/s^2$. (Neglect the effect of atmospheric pressure).



11. A uniform rod has mass m = 2 kg and length *l* = 13m. One end of the rod is pulled with a constant velocity of v_0 = 34 m/s along a frictionless horizontal floor in the negative *x* direction. The other end is moving along a parabolic fixed curve. The equation of the parabola is $x^2 = 20y$. Find the angular velocity of the rod (in rad/s) when the end point 'B' is at (10,5)



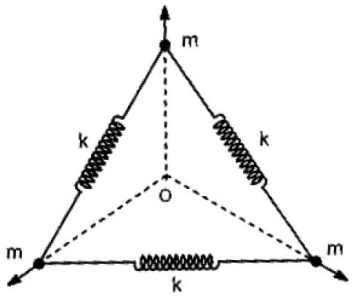
12. In the given A.C circuit the r.m.s voltage of the source is 60 V then the r.m.s current passing through the source is $(10+x)A$ then the value of *x* is



SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. Three identical springs each of force constant k have been joined to the three identical balls (each of mass m), as shown in the figure which are in at the three vertices of an equilateral triangle. In the shown arrangement, each of the spring is in its natural length. What all three balls are simultaneously given small displacements of equal magnitude along the directions as shown in the figure, the oscillation frequency for the blocks will be $\frac{1}{2\pi}\sqrt{\frac{\beta k}{2m}}$ then β =



14. An electron and proton are separated by a large distance and the electron approaches the proton with an initial kinetic energy of 2.9 eV. The electron is captured by the proton to form a hydrogen atom in the ground state. If the wavelength (in nm) of the photon given off is 15 X nm, find the value of X.

15. A neutral particle of mass 28g is at rest in a uniform magnetic field $\text{B}=\frac{200}{3}\text{T}$. At $\text{t}=0$, particle decays into two particle of equal mass and one of them having charge $3\mu\text{C}$. Both of them move in separate paths lying in the plane perpendicular to magnetic field B. At later time t, the particle collides. (neglecting the interaction force). Calculate $\frac{\text{t}}{100}$.

16. A vertical capillary tube with inside radius 0.25 mm is submerged into water so that the length of its part protruding over the water surface is equal to 25 mm. Surface tension of water is $73\times10^{-3}\,\text{N/m}$ and angle of contact is zero degree for glass and water, acceleration due to gravity is $9.8\,\text{m/s}^2$. Then value of 10R approximately (in mm) is (where R is radius of meniscus and h is height of water in capillary tube)

17. A wire of length L is shaped into a circle and then bent in such a way that the two semi-circles are perpendicular. The magnetic moment of the system when current flows through the system is $\left(\frac{\sqrt{2}\text{iL}^2}{8\pi}\right)\text{n}$, then value of n is

18. If the sound heard by observer, whose equation is given as $y=8\sin10\pi t\cos200\pi t$ at $x=0$, then number of beat frequency heard by observer is 2k. Then the value of k is

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Which of the following options is/are correct regarding the XeO_3F_2 and $XeOF_4$ molecules?
    (A) Bond length $d_{Xe-O(XeO_3F_2)} < d_{Xe-O(XeOF_4)}$
    (B) Bond length $d_{Xe-F(XeO_3F_2)} < d_{Xe-F(XeOF_4)}$
    (C) Shapes are trigonal bipyramidal and square pyramidal, respectively.
    (D) The nature of $\pi-$ bond is of $5d_\pi - 2p_\pi$ type.

20. Carbon reduction cannot be employed for the extraction of Al from Al_2O_3 because____
    (A) The temperature requirement is very high $(\sim 2000°C)$
    (B) The Ellingham diagram says Al_2O_3 is impossible to reduce by the carbon reduction process.
    (C) Al obtained in the vapour state is very difficult to handle.
    (D) Al forms its carbide (Al_4C_3) and a lot of it gets wasted.

21. Which of the following statements are correct with regard to the borax bead test?
    (A) From the result of the borax bead test, the true oxidation state of the metal cannot be predicted.
    (B) This test is applicable for all compounds.
    (C) This test is applicable for coloured compounds.
    (D) The composition of the bead produced is $NaBO_2 + B_2O_3$.

22. Which of the following statements are correct regarding the diagonal relationship between Al and Be?
    (A) BeO and Al_2O_3 are amphoteric in nature.
    (B) Carbides of both produce the same gas on hydrolysis.
    (C) Both can form complexes.
    (D) Hydrides of both the elements are covalent in nature.

23. In which of the following cases, the element is oxidized into its highest oxidation state when it reacted with O3 under moist conditions?
    (A) S
    (B) I_2
    (C) P
    (D) As

24. Which of the following pairs of compounds has the same EAN value but does not obey Sidgwick EAN rule?
    (A) $Fe(CO)_5$, $Ni(CO)_4$
    (B) $[Fe(NH_3)_6]^{2+}$, $[Cr(C_2O_4)_3]^{3-}$

(C) $\left[\mathrm{Fe}\left(H_2O\right)_5\,NO\right]^{2+}, \left[Mn\left(CO\right)_6\right]$

(D) $\left[Ti\left(CO\right)_6\right], \left[Mn\left(C_2O_4\right)_3\right]^{3-}$

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

25. The maximum number of atoms that may lie in the same plane in $P(CH_3)_3\ (CF_3)_2$ is _____x_____ and the number of such planes is _____y_____. Then find x-y

26. Find the number of type of metals present in a mixture of ore containing sylvinite, Indian saltpetre, colemanite, dolomite and anglesite.

27. How many of the following reagents can be used to distinguish between SO_2 and CO_2?
$\mathrm{Ca(OH)_2}$, $(\text{starch} + \mathrm{KIO_3})$, $(\mathrm{K_2Cr_2O_7} + \mathrm{H_2SO_4})$, $(\mathrm{KMnO_4} + \mathrm{H_2SO_4})$, $\mathrm{Ba(OH)_2}$, $\mathrm{H_2O_2}$, $\mathrm{FeCl_3}$, $\mathrm{Zn\left[Fe(CN)_5(NO)\right]}$

28. Among the following the number of pairs of compound, for which the thermal stability order is correct is_____.
(i) $\mathrm{BeCO_3 > SrCO_3}$
(ii) $\mathrm{MgO > BaO}$
(iii) $\mathrm{Li_2CO_3 < Cs_2CO_3}$
(iv) $\mathrm{CaSO_4 < BaSO_4}$
(v) $\mathrm{Li_3N > Na_3N}$
(vi) $\mathrm{LiClO_4 < KClO_4}$

29. How many statements is/are correct
(i) Monovalent silver complexes (coordination number 2) are diamagnetic.
(ii) Bivalent silver complexes (coordination number 4 and 6) are paramagnetic with μ = 1.73 BM.
(iii) $[MnBr_4]^{2-}$ ion is paramagnetic and sp^3 hybridized.
(iv) $\mathrm{Ag_2O}$ is diamagnetic.
(v) In ferrocene Cyclopentadienyl act as a π-donor ligand.
(vi) In ferrocene Fe is in (+2) oxidation state.
(vii) Ferrocene is organometallic compound.
(viii) Ferrocene has a metal–carbon bond
(ix) $Ni(CO)_4$ is diamagnetic and sp3 hybridized.
(x) Complex $[Cu(CN)_4]^{3-}$ is paramagnetic and Cu^{2+} ion is dsp^2 hybridized.

30. For $\left[\mathrm{CoCl_n\left(NH_3\right)_{6-n}}\right]\mathrm{Cl_{3-n}}$. The value of n is such that the complex cannot show optical isomerism? Then sum of such values of n is

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. The rate constants for two parallel reactions were found to be $1\times10^{-2}\,dm^{-3}mol^{-1}\sec^{-1}$ and $3.0\times10^{-2}\,dm^{-3}mol^{-1}\sec^{-1}$. If the corresponding energies of activation of parallel reaction are 60 kJ mol^{-1} and 70 kJ mol^{-1} respectively. The apparent overall activation energy of reactions is_____

32. When $\mathrm{PCl_5(g)}(M.W=208.2)$ is heated and if dissociates into $\mathrm{PCl_3(g)}$ and $\mathrm{Cl_2(g)}$. The densities of the gas mixture at 200°C and 250°C are 60.2 and 57 g/cc respectively at equilibrium. Then the degree of dissociation of $\mathrm{PCl_5}$ at $250^\circ\mathrm{C}$. Accurate upto 2 decimal places after round off.

33. A vessel contains dimethyl ether at a pressure of 0.4 atm. Dimethyl ether decomposes as $\mathrm{CH_3OCH_3(g)}\rightarrow CH_4(g)+CO(g)+H_2(g)$. The rate constant of decomposition is $4.78\times10^{-3}\,\mathrm{min^{-1}}$. Calculate the ratio of initial rate of diffusion to rate of diffusion after 4.5 hr of initiation of decomposition. Assume the composition of gas present and composition of gas diffusing to the same. (Given log 3.5 = 0.56)

34. A very small amount of a non-volatile solute (that does no dissociate) is dissolved in $56.8\,\mathrm{cm^3}$ of benzene (density $0.89\,\mathrm{g\,cm^{-3}}$). At room temperature, vapour pressure of this solution is 98.88 mm Hg while that of benzene is 100 mm Hg. If the freezing temperature of this solution is 0.73 degree lower than that of benzene, what is the value of molal freezing depression constant of benzene in nearest integer?

35. Mg can reduce $\mathrm{NO_3^-}$ to $\mathrm{NH_3}$ in basic medium.

$\mathrm{NO_3^-+Mg(s)+H_2O}\rightarrow Mg(OH)_2(s)+OH^-(aq.)+NH_3(g)$

A 25.0 mL sample of $\mathrm{NO_3^-}$ solution was treated with Mg. The $\mathrm{NH_3(g)}$ was passed into $100\,\mathrm{mL}$ of $0.15\mathrm{N\ HCl}$. The excess of HCl required $32.10\mathrm{mL}$ of $0.10\mathrm{N\ NaOH}$ for neutralisation. What was the molarity of $\mathrm{NO_3^-}$ ions in the original sample? Give your answer after rounding off to two decimal places.

36. Methane (Considered to be an ideal gas) initially at 25°C and 1 bar pressure is heated at constant pressure until the volume has doubled. The variation of the molar heat capacity with absolute temperature is given by:

$\mathrm{C_p}=22.34+48.1\times10^{-3}T$, where $\mathrm{C_p}$ is in $\mathrm{JK^{-1}\ mol^{-1}}$. Calculate molar $\Delta\mathrm{U}$. Give your answer after rounding off to two decimal places.

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If roots of quadratic polynomial $f(x) = a^2x^2 - 8ax + 7, a \in R - \{0\}$ lies on either side of unity, then the possible value(s) of a is/are:
(A) 1
(B) 3
(C) 5
(D) 7

38. If $w = \frac{z-1}{z-2}, |z| = 1, z, w \in C$, then:
(A) $|w-3|_{\min} = \frac{7}{3}$
(B) $|w-3|_{\max} = 3$
(C) $(Arg(w))_{\max} = \frac{\pi}{4}(-\pi < Arg(w) < \pi)$
(D) $(Arg(w))_{\min} = 0(-\pi < Arg(w) < \pi)$

39. f: R $\to$ R; f(x) = sgn{sin x}; where {.} denotes the fractional part function, then
(A) f(x) is a periodic function
(B) f(x) is many-one function
(C) f(x) is an even function
(D) f(x) is neither even nor odd

40. Which of the following statements is (are) true?
(A) $\int_{2007}^{2009} (\ell nx)^{2007} dx < \int_{2007}^{2009} (\ell nx)^{2009} dx$
(B) $\int_0^1 f(x)dx < \int_0^1 xf(x)dx$ where $f(x) > 0 \ \forall \ x \in [0, 1]$
(C) $\int_x^1 \frac{dt}{1+t^2} = \int_1^{1/x} \frac{dt}{1+t^2} (x > 0)$
(D) $\int_0^\infty e^{-x^2} dx = \int_0^\infty \frac{e^{-x}}{\sqrt{x}} \ dx$

41. If f(x) and g(x) are functions such that f(x + y) = f(x) . g(y) + g(x) . f(y), then $\begin{vmatrix} f(\alpha) & g(\alpha) & f(\alpha+\theta) \\ f(\beta) & g(\beta) & f(\beta+\theta) \\ f(\gamma) & g(\gamma) & f(\gamma+\theta) \end{vmatrix}$ is independent of
(A) α
(B) β
(C) γ
(D) θ

42. Tangents are drawn from (–2, 0) to $y^2 = 8x$, radius of circle(s) that would touch these tangents and the corresponding chord of contact, can be equal to

(A) $4\left(\sqrt{2}+1\right)$

(B) $4\left(\sqrt{2}-1\right)$

(C) $8\sqrt{2}$

(D) None of these

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

43. Let $f\left(x\right)=2\cot x+\left(2a+1\right)\ln\left|\cos ec\,x\right|+\left(2-a\right)x$ is strictly decreasing in $\left(0,\frac{\pi}{2}\right)$. If range of '$a$' is $\left[m,\infty\right)$, then the value of m is_____

44. If $f\left(x\right)=x^3+3x^2+\left(4-k\right)x+b$ is an injective function $\forall\, x\in R,$ then maximum integral value of k is_____

45. The value of $\sum_{r=1}^{\infty}\frac{\left(2r-1\right)}{2^r}$ is_____

46. Let a,b and c be three distinct complex numbers satisfying $\left|a\right|=\left|b\right|=\left|c\right|=1$ and $a+b+c=0$, then $\left|a^2+b^2+c^2\right|$ is equal to:

47. The sides of a triangle are in the ratio 3 : 5 and the third side is 16, if the largest possible area of the triangle is k, then $\frac{k}{30}$ is _____

48. The area enclosed by the curve max (|x – 1|, |y|) = 4 is A, then $\frac{3A}{32}$ is equal to _____

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. If the equation $x^{14}-kx^4+x^2+2=0$ has no real solution, then the greatest possible integral value of 'k' is:

50. Let a,b,c be the roots of the equation $x^4+x^3+x^2+x-1=0$. Let $f\left(x\right)=x^6-6x^2+6x+7$ and $g\left(x\right)=px^2+qx+r\left(p,q,r\in R,p\neq 0\right)$.

If $f\left(a\right)=g\left(a\right);\ f\left(b\right)=g\left(b\right)$ and $f\left(c\right)=g\left(c\right)$; then the maximum value of $g\left(x\right)$ is:

51. If $\frac{\tan\theta}{\tan\theta-\tan 3\theta}=\frac{1}{3}$, then the value of $\frac{\cot\theta}{\cot\theta-\cot 3\theta}$ is equal to_____

52. The distance of the point of intersection of the planes $\vec{r}.\left(2\hat{i}+\hat{j}\right)=0,\ \vec{r}.\left(4\hat{i}+3\hat{k}\right)=24$ and $\vec{r}.\left(12\hat{i}-7\hat{k}\right)=24$ from the origin is_____

53. If $y=\dfrac{x}{x^2+\dfrac{x}{x^2+\dfrac{x}{x^2+.....\infty}}}$ and $\int\frac{\left(y-x^2\right)dx}{\left(x^2+y\right)\left(x+y^2\right)}=f\left(y\right)+c$ then $f'\left(y\right)$ for $x=1$ is equal to_____

54. If $b\cos x+\frac{b\cos 3x}{2}\le 3$ for all x, then the number of integral values of b is _____